

❖ Features of Air Leak Tester

Bypass circuit

Save CHG time for low pressure & large volume Work

■ Summary

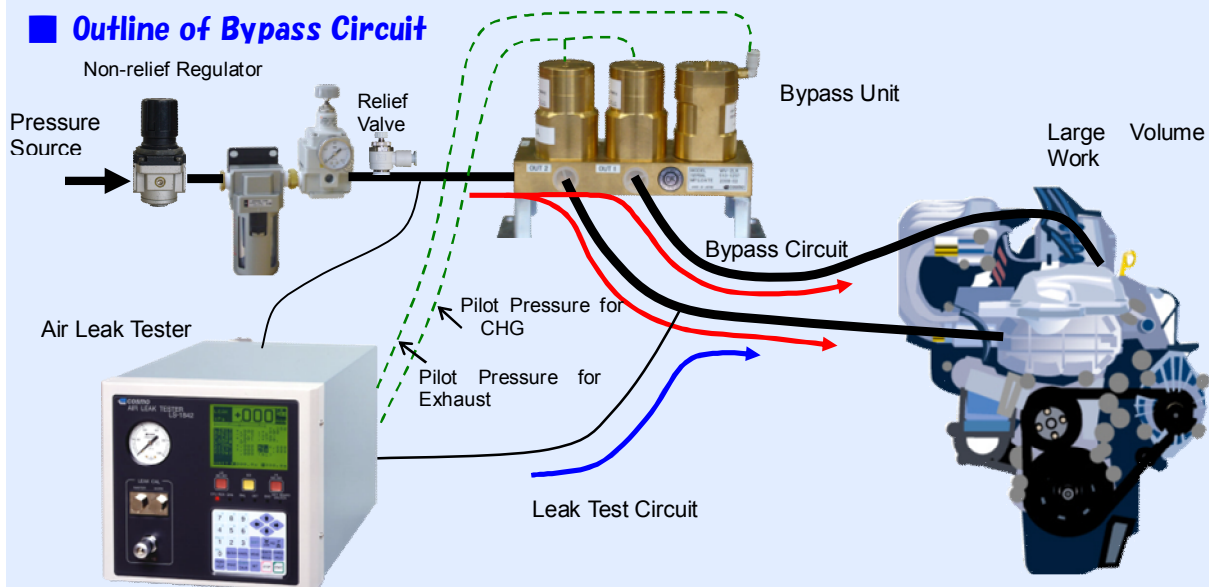
Large volume Work usually requires longer CHG time at Leak test. In that case, Bypass circuit is the ideal tool and can dramatically shorten CHG time, especially effective for a large volume Work like more than 3L with low-test pressure less than 50kPa.

Also, Bypass unit equips the exhaust valve in order not only avoiding contaminants from Work but shorten exhaust time.

■ Advantages of Bypass circuit

- **Shorten Measurement Time**
Shorten CHG time by large diameter valve
- **Shorten Exhaust Time**
Also shortens exhaust time by large diameter valve
- **Highly Reliable Bypass Valve**
Highly reliable and durable air-operated valve specialized for Leak test
- **Bypass Valve controlled by Leak Tester**
Valves of Bypass unit are controlled by pilot pressure through Leak tester

■ Outline of Bypass Circuit



■ What is Bypass Circuit???

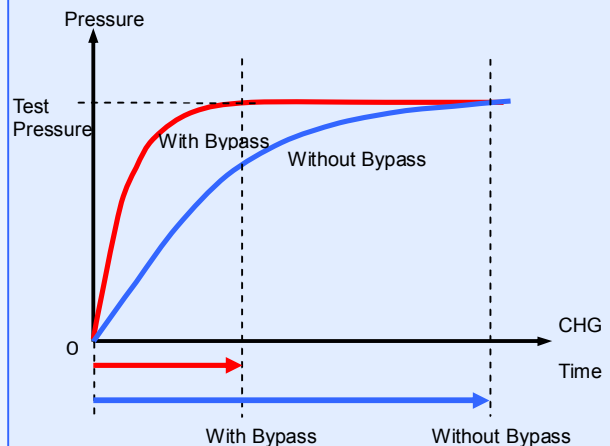
Extra route (not Leak test circuit) supplies pressure to Work with a thick tube in CHG time called Bypass circuit.

Bypass valve in Bypass circuit equips specialized valve allowing large volume of flow. Also, two or three extra fill valves allow extra route #2 or #3 in order to pressurize different location simultaneously. The more routes you have, the faster Work filled by test pressure (The above chart shows Bypass circuit with two fill valves)

Blue line on the right chart shows CHG time to fill Work without Bypass circuit. As you see, it takes longer than one with Bypass circuit (Red line).

By installing extra Bypass circuits definitely shortens measurement time.

Comparison of Pressurization with/without Bypass circuit



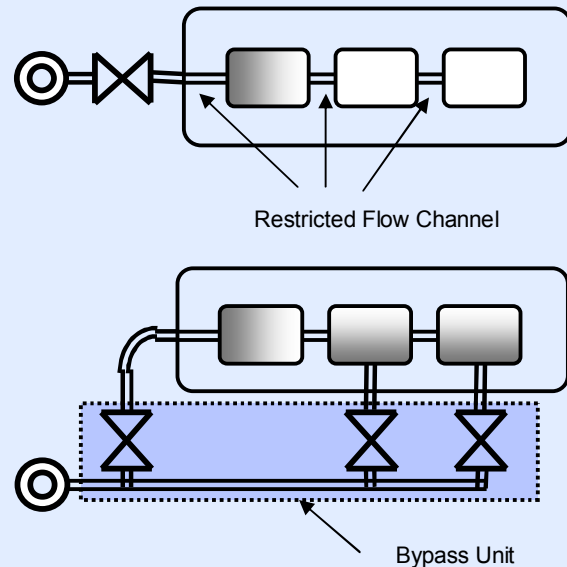
■ Complicated Internal Structure Work

◇ Problems

In some cases, CHG time takes longer to fill Work due to its structural problems such as very small inlet, several rooms in one Work, restricted flow channel etc. Moreover, it takes much longer when rooms to be filled lined up in straight position (as shown on the upper-right figure), especially this symptom more obvious when test pressure is less than 50kPa.

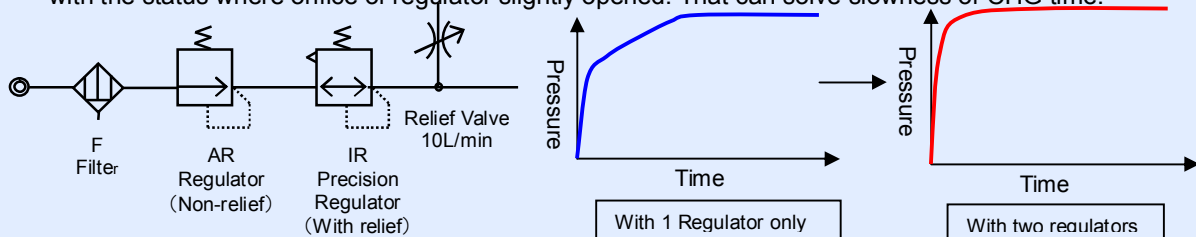
◇ Solutions

Bypass unit (as shown on the lower-right figure) allows shorter measurement time by installing individual pressurizing route. Especially effective when test pressure is less than 50kPa



■ Pressure Stabilization of Regulator

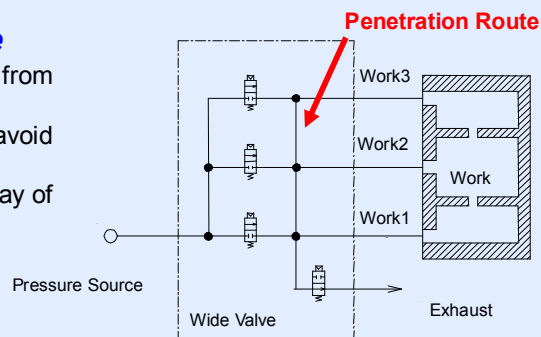
- When it comes to repeatability of pressurization, the precision regulator is superior to the electro-pneumatic one controlled with frequent interval. Even the precision regulator, however, has problems of controlling pressurization if test pressure is less than 50kpa
- In order to solve that problem, it is truly effective if installing two regulators in row and a relief valve at the end (as shown on the below figure).
- Pressure difference of the first regulator and the second one should be somewhere in between of 100 to 150kPa. For example, if test pressure is 100kPa, the first regulator recommended setting at 200kPa.
- By lowering air pressure on inlet side of the second regulator and keeping flow rate, CHG time starts with the status where orifice of regulator slightly opened. That can solve slowness of CHG time.



■ Exhaust Route Penetration K-type

Pressurized air inside of Work is exhausted from Work 1, 2, and 3.

- Exhaust from each port at the same time to avoid the delay of exhaust leak
- If Work has any leakage, it can avoid the delay of leak transmission



■ Bypass Unit

	BU-100-1	BU-100-2	BU-100-3
Number of Fill Valve	1	2	3
Rough Target of Work Volume*1	3~6L	6~20L	More than 20L
Test Pressure Range	-100kPa (Depending on Vacuum source) ~ 500kPa		

*1: Work volume is just giving you rough idea. It heavily depends on the actual environment or condition